

Bank Marketing Campaign Prediction Using Machine Learning

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Abstract—Predicting customer responses to bank marketing campaigns is an important machine learning problem that can help financial institutions improve campaign efficiency and customer targeting. This paper presents an end-to-end machine learning approach for predicting whether a customer will subscribe to a bank term deposit. The Bank Marketing dataset containing 41,188 customer records and 21 attributes was analyzed through data quality assessment, exploratory data analysis, preprocessing, feature transformation, and supervised classification.

Twelve duplicate records were identified and removed, resulting in 41,176 records. The target variable was highly imbalanced, with 88.73% of customers not subscribing and 11.27% subscribing. Categorical attributes were transformed using One-Hot Encoding, while numerical attributes were standardized using StandardScaler. The duration attribute was removed because it represents the length of the current call and is not available in a realistic pre-contact prediction scenario.

Three classification algorithms were implemented: Logistic Regression, Decision Tree, and Random Forest. The models were evaluated using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and confusion matrices. Among the evaluated models, Random Forest obtained the highest ROC-AUC of 0.8093, while Decision Tree achieved the highest Accuracy of 0.9009 and F1-Score of 0.3704. The results demonstrate the importance of evaluating multiple metrics when dealing with an imbalanced customer-response prediction problem.

Index Terms—Bank Marketing, Machine Learning, Customer Subscription Prediction, Logistic Regression, Decision Tree, Random Forest, Classification, ROC-AUC, Customer Analytics

I. INTRODUCTION

Banking institutions frequently conduct marketing campaigns to promote financial products such as term deposits. A major challenge is identifying customers who are most likely to respond positively to a campaign. Traditional marketing approaches may involve contacting a large number of customers, which can increase campaign cost while producing relatively few successful subscriptions.

Machine learning provides a data-driven approach for predicting customer responses based on demographic, financial, and campaign information. Classification algorithms can learn relationships between customer attributes and historical campaign outcomes and subsequently predict the likelihood of a new customer subscribing to a term deposit.

This project develops an end-to-end machine learning classification system using the Bank Marketing dataset. The

workflow includes data loading, data quality analysis, duplicate removal, exploratory data analysis, categorical encoding, numerical feature scaling, train-test splitting, model training, model evaluation, feature importance analysis, and prediction.

Three machine learning algorithms are considered in this study: Logistic Regression, Decision Tree, and Random Forest. Their performance is compared using multiple classification metrics.

II. PROBLEM STATEMENT

Bank marketing campaigns may require contacting a large number of customers to obtain a relatively small number of successful term deposit subscriptions. Therefore, an intelligent prediction system is required to identify customers who are more likely to subscribe.

The problem can be formulated as a binary classification task:

$$y = \begin{cases} 0, & \text{customer does not subscribe} \\ 1, & \text{customer subscribes} \end{cases} \quad (1)$$

Given a set of customer and campaign-related features X , the objective is to learn a classification function:

$$f(X) \rightarrow y \quad (2)$$

where y represents the predicted term-deposit subscription outcome.

III. RESEARCH OBJECTIVES

The major objectives of this project are:

- 1) To analyze the Bank Marketing dataset and understand its customer and campaign attributes.
- 2) To perform data quality analysis and remove duplicate observations.
- 3) To perform exploratory data analysis on numerical and categorical features.
- 4) To preprocess categorical and numerical variables for machine learning.
- 5) To develop Logistic Regression, Decision Tree, and Random Forest classification models.
- 6) To compare the models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- 7) To identify important features influencing customer subscription prediction.

IV. DATASET DESCRIPTION

The Bank Marketing dataset contains information about customers contacted during direct marketing campaigns of a banking institution. The original dataset contains 41,188 observations and 21 attributes.

The target variable is y , which indicates whether the customer subscribed to a bank term deposit.

The dataset contains both numerical and categorical variables.

A. Numerical Attributes

The numerical attributes include:

- age
- duration
- campaign
- pdays
- previous
- emp.var.rate
- cons.price.idx
- cons.conf.idx
- euribor3m
- nr.employed

B. Categorical Attributes

The categorical attributes include:

- job
- marital
- education
- default
- housing
- loan
- contact
- month
- day_of_week
- poutcome

V. DATA PREPROCESSING

Data preprocessing was performed before model development to improve data quality and convert the raw dataset into a suitable numerical representation.

A. Data Quality Analysis

The dataset initially contained 41,188 records and 21 columns. No null values were detected in the dataset.

However, 12 exact duplicate records were identified. These records were removed, resulting in 41,176 observations.

B. Handling Unknown Values

Several categorical attributes contained the value `unknown`. The number of unknown values was examined rather than automatically removing these records.

After duplicate removal, unknown values were present in attributes such as job, marital status, education, default, housing, and loan.

Since these values represent unavailable or unspecified categorical information, they were retained and handled through categorical encoding.

TABLE I
DATASET SUMMARY

Property	Value
Original Records	41,188
Original Features	21
Duplicate Records	12
Final Records	41,176
Target Classes	2
Training Records	32,940
Testing Records	8,236
Final Features	62

C. Target Variable Encoding

The target variable originally contained two string categories:

- no
- yes

These values were transformed into binary values:

$$\text{no} = 0, \quad \text{yes} = 1 \quad (3)$$

The resulting distribution was:

- Class 0: 36,537 customers (88.73%)
- Class 1: 4,639 customers (11.27%)

This demonstrates significant class imbalance.

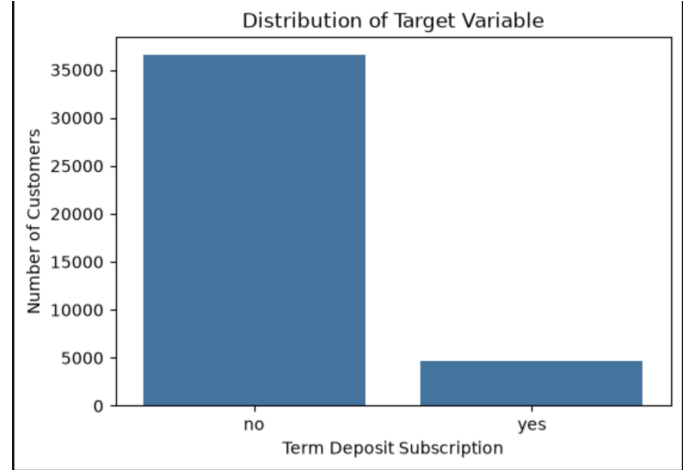


Fig. 1. Distribution of the target variable.

VI. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was performed to understand the distribution of customer attributes and their relationship with term-deposit subscription.

A. Numerical Feature Analysis

Histograms and box plots were used to examine numerical variables. The analysis provided information about feature distributions, variation, and potential outliers.

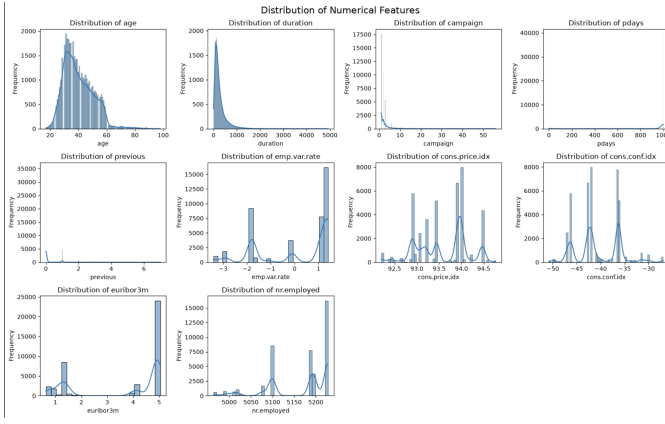


Fig. 2. Distribution of numerical features.

B. Categorical Feature Analysis

Customer attributes such as job, education, marital status, contact type, and previous campaign outcome were analyzed with respect to subscription behavior.

The analysis showed that subscription rates differ considerably between customer groups. For example, the subscription rate among students was approximately 31.43%, while the rate for blue-collar customers was approximately 6.90%.

The previous campaign outcome also demonstrated a strong relationship with subscription. Customers with a previous successful campaign outcome had a current subscription rate of approximately 65.11%.

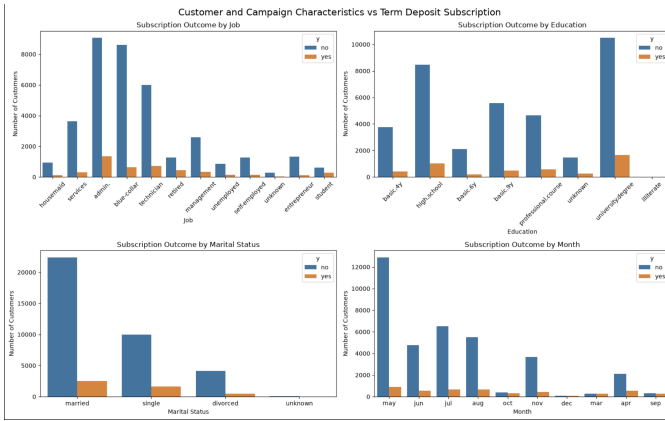


Fig. 3. Customer and campaign characteristics versus subscription.

C. Analysis of pdays

The pdays feature represents the number of days since a customer was previously contacted. A value of 999 represents customers who were not previously contacted.

Approximately 96.32% of the records contained pdays=999. This special value was retained because it contains meaningful information about previous campaign contact history.

D. Correlation Analysis

A correlation matrix was generated for the numerical features. The strongest numerical correlations with the target were observed for:

- duration: 0.4053
- previous: 0.2302
- cons.conf.idx: 0.0548
- age: 0.0304
- campaign: -0.0664
- cons.price.idx: -0.1361
- emp.var.rate: -0.2983
- euribor3m: -0.3077
- pdays: -0.3249
- nr.employed: -0.3547

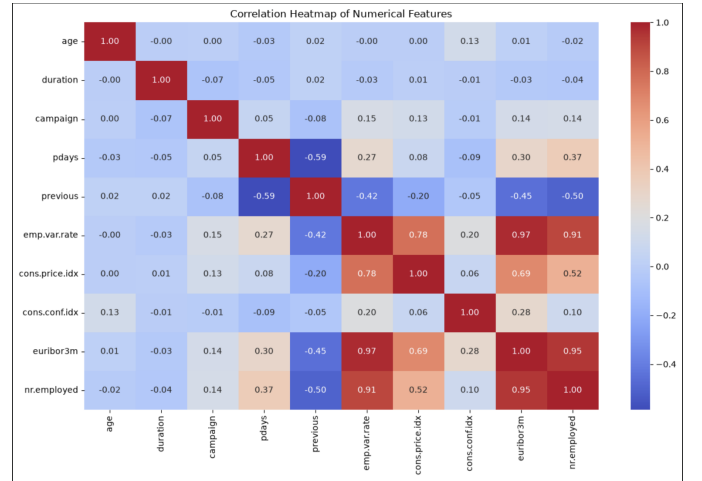


Fig. 4. Correlation heatmap of numerical features.

VII. FEATURE SELECTION

The duration attribute was excluded from model training. Although it showed a relatively strong positive relationship with the target, it represents the duration of the current marketing call.

The value of this attribute is generally known only after or during the interaction. Including it could therefore introduce information that would not be available when deciding which customer to contact before the campaign interaction.

Consequently, the final prediction system uses 19 original input features after removing the target and duration.

VIII. FEATURE TRANSFORMATION

A. One-Hot Encoding

Categorical variables were transformed using One-Hot Encoding. The encoder was configured with:

```
OneHotEncoder(
    handle_unknown="ignore",
    sparse_output=False
)
```

The categorical features produced 53 encoded features.

B. Standardization

Numerical features were standardized using StandardScaler. For a numerical feature x , the standardized value is:

$$z = \frac{x - \mu}{\sigma} \quad (4)$$

where μ is the training-set mean and σ is the training-set standard deviation.

Nine numerical features were standardized.

C. Final Feature Representation

The numerical and encoded categorical features were horizontally combined:

$$X_{final} = [X_{numerical} \mid X_{categorical}] \quad (5)$$

This produced 62 final input features.

IX. TRAIN-TEST SPLITTING

The dataset was divided into training and testing subsets using an 80:20 ratio.

Stratified sampling was used to preserve the target-class distribution in both subsets.

TABLE II
TRAIN-TEST DISTRIBUTION

	Training	Testing
Total Records	32,940	8,236
Class 0	29,229	7,308
Class 1	3,711	928

X. MACHINE LEARNING MODELS

Three supervised classification algorithms were implemented.

A. Logistic Regression

Logistic Regression estimates the probability of a binary outcome. The logistic function is:

$$P(y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}} \quad (6)$$

A probability threshold is used to convert the predicted probability into a binary class.

B. Decision Tree

Decision Tree classification recursively divides the feature space using decision rules. The model used in this project was configured with a maximum depth of 5.

Decision Trees provide an interpretable representation of classification decisions and can model nonlinear relationships.

C. Random Forest

Random Forest is an ensemble learning method that combines multiple decision trees.

The model used 100 trees with a maximum depth of 10. The final prediction is obtained by aggregating the predictions from the individual trees.

XI. SYSTEM WORKFLOW

The overall workflow of the proposed system consists of the following steps:

- 1) Load the Bank Marketing dataset.
- 2) Perform data quality analysis.
- 3) Remove duplicate records.
- 4) Analyze unknown categorical values.
- 5) Perform exploratory data analysis.
- 6) Encode the target variable.
- 7) Remove the `duration` feature.
- 8) Split the dataset into training and testing subsets.
- 9) Apply One-Hot Encoding to categorical variables.
- 10) Standardize numerical variables.
- 11) Combine transformed features.
- 12) Train Logistic Regression, Decision Tree, and Random Forest.
- 13) Generate test predictions.
- 14) Evaluate model performance.
- 15) Analyze Random Forest feature importance.
- 16) Save the final Random Forest model and preprocessing objects.

XII. EVALUATION METRICS

Since the dataset is imbalanced, multiple evaluation metrics were used instead of relying only on accuracy.

A. Accuracy

Accuracy measures the proportion of correctly classified samples:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

B. Precision

Precision measures the proportion of predicted positive samples that are actually positive:

$$Precision = \frac{TP}{TP + FP} \quad (8)$$

C. Recall

Recall measures the proportion of actual positive samples correctly identified:

$$Recall = \frac{TP}{TP + FN} \quad (9)$$

D. F1-Score

F1-Score is the harmonic mean of precision and recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (10)$$

E. ROC-AUC

ROC-AUC measures the ability of the classifier to distinguish between positive and negative classes over different classification thresholds.

TABLE III
PERFORMANCE COMPARISON OF CLASSIFICATION MODELS

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.8983	0.6500	0.2101	0.3176	0.8004
Decision Tree	0.9009	0.6522	0.2586	0.3704	0.7910
Random Forest	0.9000	0.6677	0.2231	0.3344	0.8093

XIII. EXPERIMENTAL RESULTS

The three models were evaluated using the test dataset.

The Decision Tree achieved the highest accuracy of 0.9009 and the highest recall and F1-Score among the three models. Random Forest achieved the highest precision of 0.6677 and the highest ROC-AUC of 0.8093.

Although the accuracy values of all three models are close, their recall values differ substantially. This demonstrates why accuracy alone should not be used to assess the performance of this dataset.

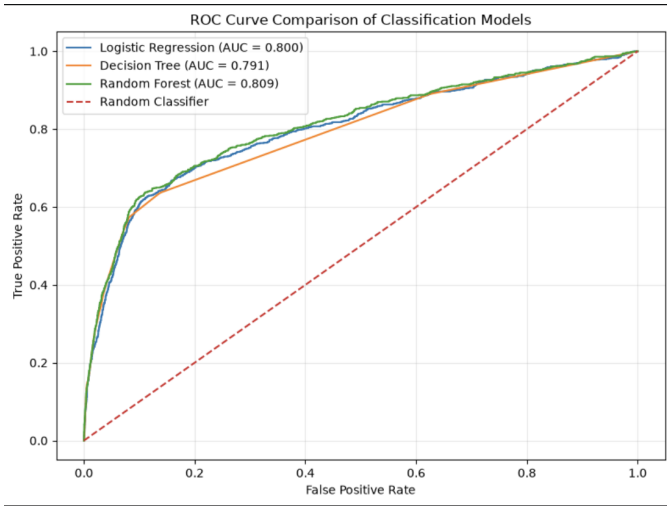


Fig. 5. ROC curve comparison of Logistic Regression, Decision Tree, and Random Forest.

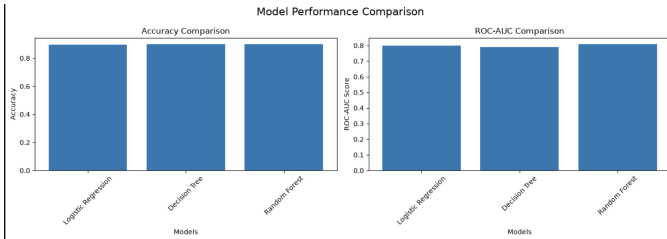


Fig. 6. Comparison of model Accuracy and ROC-AUC.

XIV. CONFUSION MATRIX ANALYSIS

The confusion matrices provide additional information about the classification behavior.

A. Logistic Regression

The Logistic Regression confusion matrix was:

$$CM_{LR} = \begin{bmatrix} 7203 & 105 \\ 733 & 195 \end{bmatrix} \quad (11)$$

The model correctly classified 7,203 negative samples and 195 positive samples.

B. Decision Tree

The Decision Tree confusion matrix was:

$$CM_{DT} = \begin{bmatrix} 7180 & 128 \\ 688 & 240 \end{bmatrix} \quad (12)$$

The model identified 240 of the 928 positive test samples.

C. Random Forest

The Random Forest confusion matrix was:

$$CM_{RF} = \begin{bmatrix} 7205 & 103 \\ 721 & 207 \end{bmatrix} \quad (13)$$

Random Forest correctly identified 207 positive samples while correctly classifying 7,205 negative samples.

XV. FEATURE IMPORTANCE ANALYSIS

Random Forest feature importance was used to identify the variables that contributed most strongly to the model's decisions.

The most important features included:

TABLE IV
TOP RANDOM FOREST FEATURES

Feature	Importance
nr.employed	0.1461
euribor3m	0.1445
poutcome_success	0.0890
pdays	0.0865
emp.var.rate	0.0809
cons.conf.idx	0.0618
cons.price.idx	0.0472
age	0.0419
previous	0.0256
campaign	0.0173

XVI. MODEL SELECTION

The Random Forest model achieved the highest ROC-AUC value of 0.8093 and the highest precision of 0.6677. Therefore, it provides the strongest overall ranking capability among the evaluated models.

However, Decision Tree achieved the highest recall and F1-Score. Therefore, the preferred model depends on the business objective.

If the goal is to maximize the ability to rank potential customers according to their predicted outcomes, Random Forest is a suitable choice. If identifying a larger proportion of actual subscribers is more important, Decision Tree may provide an advantage under the current configuration.

XVII. MODEL PERSISTENCE AND DEPLOYMENT

The final Random Forest model was saved using the Joblib library. The preprocessing components were also saved so that new input data can undergo the same transformations used during training.

The following objects were stored:

- Random Forest model
- One-Hot Encoder
- Standard Scaler

This allows the trained system to be reused for future prediction tasks without retraining the complete model.

XVIII. DISCUSSION

The experimental results demonstrate that customer subscription prediction is a challenging classification problem due to the significant imbalance between positive and negative classes.

All three models achieved approximately 90% accuracy. However, the positive-class recall remained relatively low. This occurs because a model can obtain high overall accuracy by correctly predicting the dominant negative class while missing a substantial number of positive customers.

Random Forest achieved the highest ROC-AUC of 0.8093, indicating better overall discrimination between subscribing and non-subscribing customers compared with the other evaluated models.

The exploratory analysis also identified important relationships between campaign characteristics and subscription outcomes. In particular, previous campaign success showed a substantially higher subscription rate than unsuccessful or nonexistent previous campaign outcomes.

Economic variables such as `nr.employed`, `euribor3m`, and `emp.var.rate` were among the most important Random Forest features.

XIX. LIMITATIONS

The project has several limitations.

- 1) The dataset contains a significant class imbalance.
- 2) The current models were not optimized using extensive hyperparameter search.
- 3) The prediction threshold was not specifically optimized for maximizing recall.

- 4) The `duration` feature was removed to avoid post-contact information leakage, although it has strong predictive power.
- 5) The experiments were performed on a historical dataset, and model performance may change when applied to new campaigns.

Future work can investigate class-weighted models, threshold optimization, ensemble optimization, and additional sampling strategies to improve positive-class detection.

XX. FUTURE WORK

Future improvements may include:

- Hyperparameter optimization for all classification models.
- Evaluation using cross-validation.
- Class-weighted learning to address class imbalance.
- SMOTE or other resampling techniques.
- Probability threshold optimization.
- Gradient Boosting and XGBoost comparison.
- Feature selection and dimensionality reduction.
- Deployment as a web-based prediction application.
- Continuous monitoring of model performance.

XXI. CONCLUSION

This paper presented an end-to-end machine learning system for predicting whether customers would subscribe to a bank term deposit.

The workflow included dataset loading, data quality assessment, duplicate removal, exploratory data analysis, target encoding, categorical encoding, numerical scaling, feature selection, train-test splitting, model training, evaluation, feature importance analysis, and model persistence.

Three classification algorithms were evaluated: Logistic Regression, Decision Tree, and Random Forest.

The Decision Tree achieved the highest accuracy of 90.09% and the highest F1-Score of 0.3704. Random Forest achieved the highest ROC-AUC of 0.8093 and precision of 0.6677.

The results demonstrate that accuracy alone is insufficient for evaluating this problem because the target variable is highly imbalanced. Recall, F1-Score, and ROC-AUC provide additional insight into the model's ability to identify customers who actually subscribe.

The Random Forest model and preprocessing components were saved for future deployment. Overall, the project demonstrates how machine learning can be applied to bank marketing data to support data-driven customer targeting and campaign analysis.

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